Assignment4

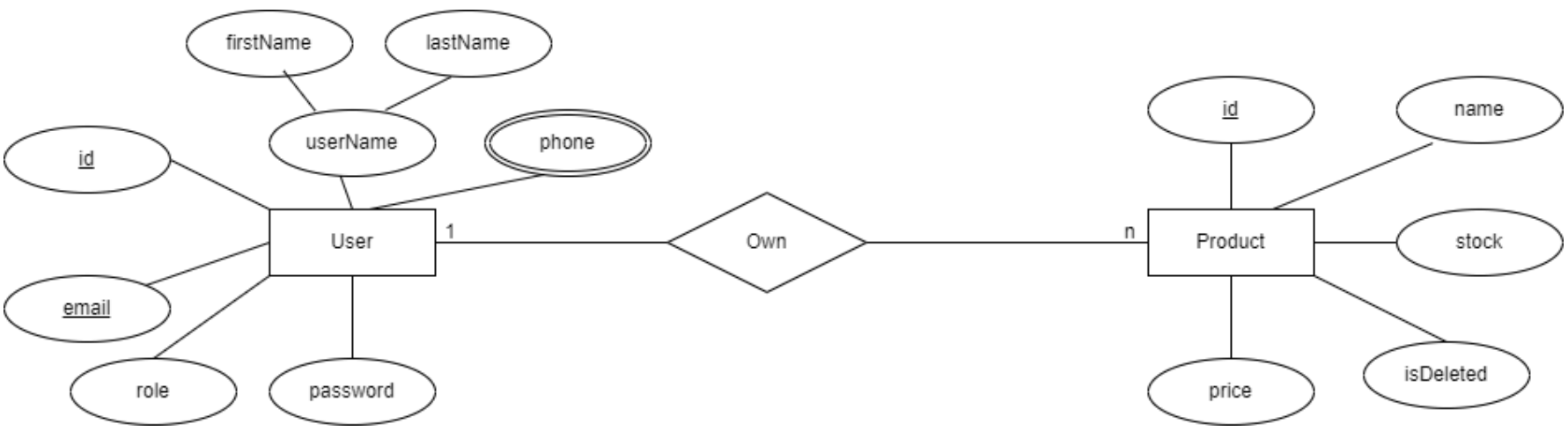
Part 1: ERD Diagram (3 Grade)

Musicana records have decided to store information on musicians who perform on their albums in a database. The company has wisely chosen to hire you as a database designer.

- Each musician that is recorded at Musicana has an ID number,a name, an address (street, city) and a phone number.
- Each instrument that is used in songs recorded at Musicana has a unique name and a musical key (e.g., C, B-flat, E-flat).
- Each album that is recorded at the Musicana label has a unique title, a copyright date, and an album identifier.
- Each song recorded at Musicana has a unique title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A producer may produce several albums.

Part2: Design a schema (Mapping) for the following ERD. (Use any design tool you want)

(2 Grade)



Part 3: (Using Node.js and MySQL) Answer the Questions below based on the given Scenario

The small retail store needs a RESTful API to manage information about its products, suppliers, and sales.

1. **Products Table:**
   - **ProductID**: Unique identifier for each product (integer, primary key, auto-increment).
   - **ProductName**: Name of the product (**text**).
   - **Price**: Price of the product (**decimal**).
   - **StockQuantity**: Quantity of the product in stock (**integer**).
   - **SupplierID**: ID of the supplier providing the product (**integer, foreign key referencing Suppliers**).
2. **Suppliers Table:**
   - **SupplierID:** Unique identifier for each product (integer, primary key, auto-increment).
   - **SupplierName**: Name of the supplier (**text**).

Assignment4

- **ContactNumber:** Supplier's contact number (**text**).

3. **Sales Table:**
   - **SaleID:** Unique identifier for each product (integer, primary key, auto-increment).
   - **ProductID:** Reference to the product sold (**integer, foreign key referencing Products**).
   - **QuantitySold:** Quantity of the product sold (**integer**).
   - **SaleDate:** Date of sale (**date**).

(Using Node.js, Express.js,and MySQL) Build a RESTful API that performs the following tasks (15 Grades):

1. Configure a MySQL **connection pool** using **mysql2/promise**, create the required database (if it does not exist), and create the **Products**, **Suppliers**, and **Sales** tables with the appropriate primary and foreign key relationships. **(0.5 Grade)**

2. Create REST API endpoints to perform **CRUD operations** for the **Products** table: **(2.5 Grade)**
   - Create a product.
   - Retrieve all products.
   - Retrieve a product by ID.
   - Update a product.
   - Delete a product.

3. Create REST API endpoints to perform **CRUD operations** for the **Suppliers** table: **(2 Grade)**
   - Create a supplier.
   - Retrieve all suppliers.
   - Update supplier information.
   - Delete a supplier.

4. Create REST API endpoints to manage **Sales** : **(1.5 Grade)**
   - Record a sale.
   - Retrieve all sales.
   - Retrieve sales for a specific product.

5. Create API endpoints to perform the following database modifications: **(2 Grade)**
   - Add a **Category** column to the **Products** table.
   - Remove the **Category** column.
   - Change **ContactNumber** to **VARCHAR(15)**.
   - Add a **NOT NULL** constraint to **ProductName**.

6. **Create an API endpoint or initialization script to insert the following data:(1.5 Grade)**
   a. Add a supplier with the name **'FreshFoods'** and contact number **'01001234567'.**
   b. Insert the following three products, all provided by **'FreshFoods'**:
      i. **'Milk'** with a price of **15.00** and stock quantity of **50.**
      ii. **'Bread'** with a price of **10.00** and stock quantity of **30.**
      iii. **'Eggs'** with a price of **20.00** and stock quantity of **40.**
   c. Add a record for the sale of **2** units of **'Milk'** made on **'2025-05-20'**.

7. Create an API endpoint to update the price of **'Bread'** to **25.00**. **(0.5 Grade)**

8. Create an API endpoint to delete the product **'Eggs'. (0.5 Grade)**

9. Create a reporting endpoint to retrieve the total quantity sold for each product using SQL aggregate functions. **(0.5 Grade)**

10. Create a reporting endpoint to retrieve the **product with the highest stock quantity**. **(0.5 Grade)**

11. Create a reporting endpoint to retrieve **suppliers whose names start with 'F'**. **(0.5 Grade)**

12. Create a reporting endpoint to retrieve **all products that have never been sold**. **(0.5 Grade)**

13. Create a reporting endpoint to retrieve **all sales** including: **(0.5 Grade)**
    - Product name
    - Quantity sold
    - Sale date using SQL **JOIN** operations.

Assignment4

14. Create a SQL script or secure administrative endpoint to create a MySQL user named **store_manager** and grant the following permissions on all tables: **(0.5 Grade)**
    - SELECT
    - INSERT
    - UPDATE

15. Revoke the **UPDATE** permission from **“store_manager”**. **(0.5 Grade)**

16. Grant **DELETE** permission to **“store_manager”** only on the Sales table. **(0.5 Grade)**

Bonus (2 Grades)

How to deliver the bonus?

1- Solve the problem Customer Who Visited but Did Not Make Any Transactions on **LeetCode**

2- Inside your assignment folder, create a **SEPARATE FILE** and name it “bonus.txt”

3- Copy the code that you have submitted on the website inside ”bonus.txt” file.